CURSO 2026-2: INTRODUCCIÓN A LA TEORÍA GEOMÉTRICA DE MEDIDAS

SEAN MCCURDY

9CP, Lunes, Miercoles, y Vienres 10:00 - 11:30am

1. Course Overview

Geometric Measure Theory (GMT) studies the deep relationship between analytic or measure-theoretic properties of an object and its geometric properties. The purpose of this class is to introduce students to the "measure-theoretic perspective" by demonstrating how GMT studies the most fundamental objects in Real Analysis: subsets of Euclidean space, Lipschitz functions, Sobolev functions, and Functions of Bounded Variation.

The course will be designed to be useful for anyone interested in Probability, PDEs, or the Calculus of Variations. It will give the necessary foundation for students who wish to study Geometric Variational Problems (isoperimetric problems, minimal surfaces, etc.), Free Boundary Problems, and geometric topics in PDEs (structure of singular sets, interior and boundary Unique continuation, structure of nodal sets, etc.).

The key themes of this course will be:

(1) The measure-theoretic function concept and its consequences. The measure-theoretic concept of a function gave rise to the notion of duality, Functional Analysis, weak derivatives, Sobolev spaces, and distribution theory. We will focus on how this measure-theoretic perspective leads to the Banach space of vector-valued Radon measures, Sobolev functions, Rellich compactness, and Functions of Bounded Variation.
(2) The Density Ratio and density functions. The study of density ratios is a foundational part of GMT which gives many important results about Sets, Lipschitz functions, Sobolev Functions, Functions of Bounded Variation. (See Topics, below.)
(3) Tangent measures. Tangent measures are the measure-theoretic version of a derivative or tangent. They are a powerful tool, and they play an important role in the study of singular sets in PDEs and Minimal Surface theory, Free Boundary problems,etc..

Additionally, homework for the course will involve many applications of these themes and the material of the course to a wide variety of important topics in Analysis, PDEs, Geometry, and the Calculus of Variations.

2. Topics

(1) Basic properties of vector-valued Radon measures as a Banach Space
   - Vector-valued measures and approximation properties of Radon measures.
   - The Riesz Representation Theorem, the Total Variation norm,
   - Weak compactness of Radon measures, weak convergence of Radon measures. Other types of convergence for measures.
   - Applications to sets, L^p functions, etc.

(2) The Vitali property of Radon measures and its immediate consequences.
   - Vitali covering lemma, Besicovitch covering lemma.
   - Density result #1: Differentiation of vector-valued Radon measures.
   - Density result #2: Lebesgue points and density theorems. Density and dimension.
   - Density result #3: Disintegration of measures (reverse Fubini for measures)

(3) The study of Lipschitz functions.
   - Extension theorems.
   - Density result #5: Rademacher's theorem, the existence of classical derivatives $\mathcal{H}^n$-a.e. for Lipschitz functions.
   - Rellich compactness.
   - Density result #6: Area and Coarea formulas (change of variables and nonlinear Fubini-type results for Lipschitz functions).

(4) The study of Sobolev functions
   - Extension results.
   - Sobolev-Poincare Inequalities. Morrey Inequality. Empbedding theorems.
   - Traces and non-tangential approximate limits of Sobolev functions.
   - Density result #7: p-capacity and fine properties of Sobolev functions-

(5) The study of Functions of Bounded Variation and sets of finite perimeter
   - Basic properties.
   - Sets of finite perimeter
   - Isoperimetric Inequalities.
   - Flemming-Rishel Coarea formula
   - Tangent measures
   - Tangent measures result #1: De Giorgi's Structure Theorem for Sets of Finite Perimeter
   - Tangent measures result #2: Characterization of the Jump set

3. Prerequisites

THIS COURSE WILL BE TAUGHT IN ENGLISH. It will be assumed that students have seen the construction of the Lebesgue integral. It is encouraged that students will have completed the Curso Bàsico: Análisis Real 1, but it is not required.

4. Bibliography

The main references will be:

(1) L. C. Evans and R. F. Gariepy, *Measure theory and fine properties of functions*, second edition of the 2015 revised edition [3409135], Textbooks in Mathematics, CRC Press, Boca Raton, FL, 2025; MR4907811
(2) L. Ambrosio, N. Fusco and D. Pallara, *Functions of bounded variation and free discontinuity problems*, Oxford Mathematical Monographs, Oxford Univ. Press, New York, 2000; MR1857292

Instituto de Matemáticas, Universidad Nacional Autonoma de México, Ciudad de México, México

Email address: sean.mccurdy@im.unam.mx